Aim : To study and Implement scikit learn library

Introduction :

This practical assignment demonstrates the step-by-step process of training a basic machine learning model using Python's scikit-learn (sklearn) library. The key phases covered in this exercise include:

Data Preparation: The practical utilizes the numpy library to generate a sample dataset, creating a feature array X consisting of 10 rows and 2 columns, alongside a target label array y with values ranging from 0 to 9.

Train-Test Splitting: The data is divided into training and testing sets using the train_test_split function. By setting the test_size to 33% (0.33), the dataset is split so that 6 rows are used for training and 4 rows are reserved for testing.

Model Initialization and Fitting: A LinearRegression model is initiated. The model is then trained (fitted) using the Model.fit() function applied to the training data (X_train and y_train).

Prediction and Evaluation: The trained model predicts outcomes for the testing data using Model.predict(X_test). The results showed that while the model correctly predicted the values 1, 2, and 3, it produced a slight mathematical error for the expected '0' value, outputting a value close to 3.44e-15 instead.

Additional Functions: The practical also explores auxiliary model functions like Model.predict_proba(), which returns the probability that an estimator has for categorical variables, and Model.score(), which evaluates the model's fit on a scale of 0 to 1, where a larger value indicates a better fit.

Importing LinearRegression from sklearn

```
from sklearn.linear_model import LinearRegression
```

Initiating Model with LinearRegression

```
Model = LinearRegression()
Model
```

```
▾ LinearRegression
LinearRegression()
```

Since sklearn is based upon Numpy and Pandas and we wanted to use sklearn's arange function

```
import numpy as np
from sklearn.model_selection import train_test_split
X, y = np.arange(20).reshape((10, 2)), range(10)
X
```

```
array([[ 0,  1],
       [ 2,  3],
       [ 4,  5],
       [ 6,  7],
       [ 8,  9],
       [10, 11],
       [12, 13],
       [14, 15],
       [16, 17],
       [18, 19]])
```

The values of Y that will 0 to 9 which are label to the array above respectively

```
list(y)
```

```
[0, 1, 2, 3, 4, 5, 6, 7, 8, 9]
```

Training , Testing and spliting

```
X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.33 ,randc
```

As we have taken 33 % for testing that is 0.33 we have left with six for train which are been displayed below

```
X_train
```

```
array([[12, 13],
       [ 8,  9],
       [14, 15],
       [16, 17],
       [18, 19],
       [10, 11]])
```

```
y_train
```

```
[6, 4, 7, 8, 9, 5]
```

Now lets see testing Data

```
X_test
```

```
array([[0, 1],
       [2, 3],
       [4, 5],
       [6, 7]])
```

```
y_test
```

```
[0, 1, 2, 3]
```

Starting with the Training process

Now we will be spliting the train and test with no shuffle therefore took y as argument with shuffle false

```
train_test_split(y, shuffle=False)
```

```
[[0, 1, 2, 3, 4, 5, 6], [7, 8, 9]]
```

Fiting the mode

```
Model.fit(X_train, y_train)
```

```
▾ LinearRegression
LinearRegression()
```

Prediction Phase

```
Mahekpredictions = Model.predict(X_test)
```

here when we checked the predicton the 1 2 3 are right just the first prediction where we expect 0 we got that 3.4416 ....

Why we got wrong prediction as we were having less rows for training and testing

In general we whenever we want to apply a technique we require a min of atleast 50 rows

 # Two more commands in the video

Model.predict_proba() : returns the probability that estimator has each categorical varibale

Model.score() : for classification and regression problem most estimators implement a score method Value is in between 0 and 1 Larger the value better fit

Conclusion

This assignment successfully illustrates the core pipeline of predictive modeling, from splitting datasets to training and extracting predictions using a Linear Regression model. The minor error observed during the prediction phase served as an important learning outcome regarding dataset limitations. It highlights that small sample sizes can lead to inaccurate predictions, reinforcing the general rule that applying such machine learning techniques typically requires a minimum of at least 50 rows of data to ensure reliability and accuracy.